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Low Cost, Portable ECG Monitoring and Alarming System Based on Deep Learning

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***Abstract*—Electrocardiogram (ECG) has been the golden standard for the detection of cardiovascular disease for many years. Any electrical impulse disruption that causes the heart to the contract may lead to arrhythmia. Arrhythmia patients have no indications of having an arrhythmia, but a doctor may recognize arrhythmias in a routine test. Therefore, continuous wearable personal monitoring system plays a big role, and it's become popular day by day. This research focuses on designing and developing a method for predicting arrhythmia (atrial fibrillation) along with monitoring the ECG signals. To create an arrhythmia prediction model and an Android-based real-time ECG surveillance system, Long Short-Term Memories neural network, Recurrent Neural Network, TensorFlow and Keras library are applied here. Those deep learning models and algorithms help to achieve overall 97.57% accuracy on arrhythmia prediction. The system is being designed with Raspberry pi 3, Arduino UNO, AD8232 single lead ECG sensor, HC-05 Bluetooth, biomedical sensor pad and battery. This system will make easier for doctors to monitor the ECG of their patients outside the hospital and also help for remote ECG monitoring. The total components cost of this research work is around USD 58.**

Keywords— arrhythmia, ECG, LSTM, RNN, raspberry pi 3, atrial fibrillation

I. Introduction

Cardiac diseases are the leading reason for death, which is 23% of the entire death in the United States in 2017 [1]. Cardiac death is mostly seen in elderly persons particularly in male adults. An estimated 180,000 to 300,000 sudden cardiac deaths occur in the US annually [2]. It is predicted that about 40% to 50% of all cardiovascular deaths are sudden cardiac deaths and most of these are caused by atrial fibrillation arrhythmia [3]. Prevention of cardiac disease diagnosis and treatment is the most significant and challenging problem for modern medicine.

Arrhythmia is a problem with the human heartbeat rhythm. This means that the heart beats too quickly, too slowly, or with an irregular pattern. Atrial fibrillation arrhythmia is one of the common arrhythmias. At atrial fibrillation, irregular electrical signal generates at the upper atria. Therefore, P wave in ECG is irregular and heartbeats go over 200 BPM [4]. For the detection of atrial fibrillation arrhythmia, the cardiologist uses an electrocardiogram that represents the electrical function of the core. An electrocardiograph is the most common test for heart conditions. The electrocardiogram is a graphic record of electrocardiography. The ECG curve contains P, Q, R, S, T and occasionally U waves. Very significant intervals and sections between waves are used to describe ECG [5]. For diagnostics of various cardiac diseases like atrial fibrillation arrhythmia, the most significant segments of an ECG signal are the QRS complex, ST segment, PR interval, RR interval, PR segment, QT interval [6].

Some patients do not have any symptoms of atrial fibrillation arrhythmia, but the doctor may be able to detect arrhythmia during a routine ECG examination [7]. It is difficult to detect arrhythmias in a short time window of the ECG waveform. Therefore, continuous monitoring of patients' heartbeats in daily life is crucial to arrhythmia detection. The quicker the diagnosis, the quicker the lives of the patient can be saved.

The design and development of this research work focus on the system architecture of detecting arrhythmia (atrial fibrillation) and alarm processing. The input signal of the architecture is obtained from the electrocardiogram. To construct an effective model for predicting arrhythmia, Long Short-Term Memory technique uses here to handle raw information from the ECG signal. This model does not require explicit preprocessing, but it can find hidden structures of various ECG entities and learn their dependencies automatically by deep machine learning. The ECG is collected and monitored in real-time within the system. Because of the sophisticated features and open-source nature, the Android smart-phone is chosen as a portable monitoring device. The reliability and time delay of this research has been tested in a real environment.

II. Related Work

A few studies have previously conducted on the use of smartphones in ECG monitoring and arrhythmia forecast. ECG categorization algorithm using Long Short-Term Memory (LSTM) reduces computational cost and time for wearable devices [8]. The deep-learning model also helps to surge the classification accuracy. For the amplified processing period, the ECG data sampling time decreases to 50%. Therefore, the accuracy of the system similarly decreases in a tolerable range. The system has no real-time ECG capturing feature.

The amalgamation of convolution Neural Networks and LSTM help to predict atrial fibrillation easily [9]. ECG data is recorded for 9 seconds to 60 seconds then the data process. This isn't a real-time data processing system. Therefore, information losses and overfitting problems remain seen in this system. In the other work [10], the authors design and develop an ECG monitoring and alarming system based on Android smartphones. An Android phone detects alarms that come from the ECG device. The smartphone will capture the

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ECG graph signals image and the specifics about the alarms and sends them to the cloud alarm server. But this system does not have any predictive characteristics of heart diseases that can help the doctor to make a specific diagnosis at a crucial time for the patients.

Wavelets and Multi-Resolution analysis base on computer-assisted arrhythmia detection help to increase the overall accuracy of 99.66% [11]. But the process can't use as a remote monitoring device because its computer connected, also a costly and complex device. Fuzzy logic with MATLAB reduces the complexity of the arrhythmia detection process [12]. Despite the complexity, the system accuracy reduces to 90%. Savitzky–Golay filter is used in MATLAB for noise cancelation but it has a low noise cancelation rate.

Early detection of atrial fibrillation using Linear Discriminant Analysis (LDA) has a low accuracy of 87% [13]. Because the LDA system reduces the sampling frequency for reducing process time, thus the ECG signal quality decrease at a significant level. A portable ECG plotting module using a microcontroller implements by authors in [14]. This module is cheap (USD 53.25) for using a microcontroller but the precious noise cancelation impossible for low processing power. There is no arrhythmia detection system, ECG data storage system and alarming system. The Bluemix cloud base ECG remote monitoring system developed by the author in [15]. Raspberry Pi 2 use for ECG data processing. Real-time data plotting has a time delay for a cloud system. The proposed system hardware cost is USD 75.88.

III. Methodology

A. Model Architecture

AD8232 single lead ECG sensor senses the voltage of the heart bit through the three biomedical sensor pads. These ECG signals are received by Arduino UNO and send to Raspberry Pi 3 for signal processing. The processed signal sends back to Arduino UNO then it sends to the mobile application using the HC-05 Bluetooth module for real-time ECG plotting. A 5V DC battery is used to power all the equipment.

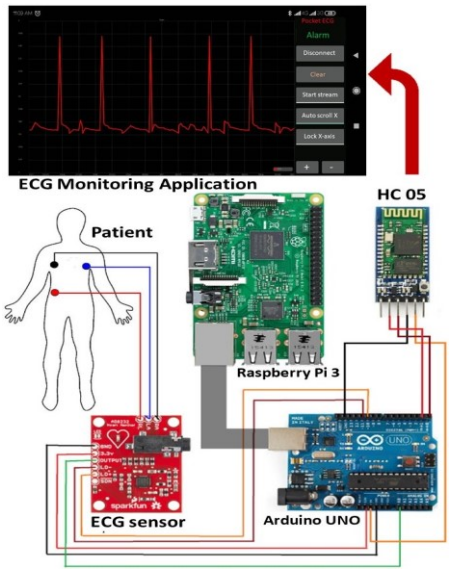

Fig. 1. Complete model architecture

B. Datasets

MIT-BIH arrhythmia ECG dataset uses for analysis of ECG signals and detecting atrial fibrillation arrhythmia. This dataset has a wide range of ECG signal samples for machine learning algorithms. There are two-channel ECG recordings of 47 patients, each approximately 30 minutes long. The records include 360 Hz sampling frequency with 11 bits resolution at 10 mV voltage range.

C. Data Preprocessing

Raspberry Pi 3 serially reads the ECG signals through Arduino by using an AD8232 Heart Rate Monitor sensor's 3 electrodes which are placed in patients' bodies seen in Fig. 4. The Raspberry Pi will process the signals and deploy the deep learning model. The deep learning model is written in Python programming language by using Keras library with a TensorFlow backend. Tensor-Flow uses dataflow charts to characterize the calculation, the shared condition, and the activities that modify that state. Keras helps to optimize the ECG data. All the signals were transformed into a vector system. A one-hot encoding technique has been used here for the conversion. In the preprocessing library, Scikit-learn includes the One Hot Encoding algorithm.

D. Long Short-Term Memory as a Classification Method

Long short-term memory (LSTM) is a deep learning structure for forecasting artificial model pathways. The system can learn the sequence data similar to the ECG signal. LSTM is a category of a recurrent network where node performs are calculated with the input of ECG signal and return output as an arrhythmia prediction. The process is feed-forwarding to the next layer's nodes. A node along with a bunch of parameters and these parameters are learned through training. The nodes are recurrent but they similarly have an inner state. The node uses the inner state as a working memory space. The recurrent architecture uses back-propagation that flows through time. LSTM prevents back-propagated errors from vanishing error data. LSTM nodes are getting complicated for huge data of ECG signals. But this makes the structure better at learning the complex real-time ECG signal data.

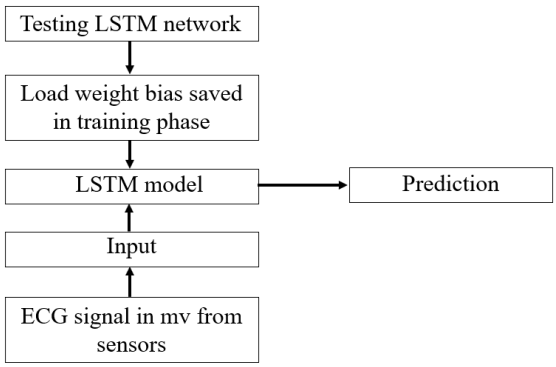

Fig. 2. Working principle block diagram of the LSTM model

E. Train the Recurrent Neural Networks Model

Back-propagation is a renowned technique for training feed-forward neural networks which cannot be practical to train the Recurrent Neural Networks (RNN) because of the current chronological dependencies in the model. Training information is created by merging two information sets, local ECG data, and worldwide ECG information. Local

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information is patient-specific and helps to increase classification precision owing to current similarities between each patient's heartbeats. In the second part, world information consists of several heartbeats from all classes of arrhythmia. It enables the model to know other patterns of arrhythmia that are not part of the local information. Train information is divided into batches of several heartbeats each to train RNN models. The heartbeats are processed as follows sequentially. On the accomplishment of the individual batch, the weights will be restructured. At the start of every load, the hidden state is set to zero and the cell state is randomly set. The input information is then spread across the network and the batch ends with error calculation. Then, the weight conditions transform and their average weight is fixed in the restructured weight, over the unfolded network. This is continued until each batch processed. This technique is identified as the back-propagation process through time.

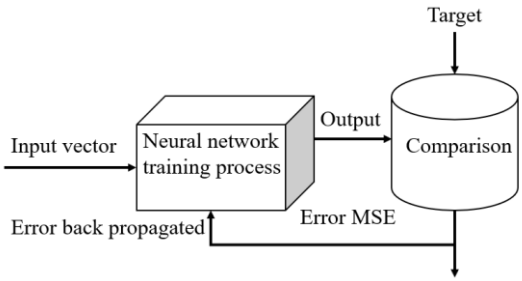

Fig. 3. Basic block diagram of the recurrent neural network

This system continues for several forecasts. Each prediction is generated where back-propagation use to spread the LSTM model over time. From these observations, the expected outcome is assumed as either zero or one. Zero is for regular sinus rhythm of the heartbeat and one is meant for arrhythmia. The predicted result will then refer to the Android smartphone through Bluetooth. If the arrhythmia is detected, an emergency alert will see in the mobile application.

F. Real-time ECG Monitoring in Mobile Application

The software framework has been developed with the latest Android SDK 29.0.1 (Google Inc.) in Java. The main graphical user interface (GUI) of the application generally consists of two features. The first feature is the prediction of atrial fibrillation arrhythmia, which enables the user to gain information about the beat classification and prediction result of arrhythmia from Raspberry Pi. The second feature is a real-time graph that allows the user to receive the ECG sensor values for plotting the ECG graph in graph-view. This graph-view is a GUI for plotting graphs with 6 features.

1) Alarm: When there is an abnormality in heart rhythm or arrhythmia (atrial fibrillation) detected, the alarm button will be color red otherwise it would be green.

2) Stop Stream: By pressing this button, the user can freeze the viewing portion of the streaming graph-view and analyze it statically.

3) Auto-scroll X: The auto-scroll button enables the user to stream the graph along X-axis.

4) Lock X-axis: This button will help the user to lock the graph along X-axis.

5) Plus: By pressing the plus button, the user will be able to zoom in the graph-view.

6) Minus: By pressing the minus button, the user will be able to zoom out the graph-view.

G. Data recording

A memory card is connected with Raspberry Pi 3 for recording ECG signal data and farther investing the signal. This system records the data for 24 hours. After 24 hours the recorded data will erase automatically and ready for collecting new data.

H. Experimental Setup

This system is tested on the real environment and the ECG signal is recorded for arrhythmia detection.

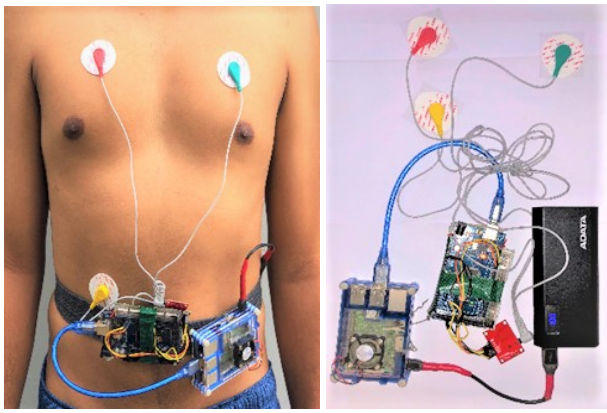

Fig. 4. Experimental setup

IV. Results and Discussion

This research work emphasis the testing and performance of ECG signal classification, precision on plotting the graph. To find the finest response, the technique of trial and error is selected for this research. The following formulation helps the research to begin. N_i the input cells, N_o the output cells, N_s the sample amount in the training information, and α is the scaling variable.

$$N_h = \frac{N_s}{(\alpha*(N_i+N_o))} \tag{1}$$

After the end of the testing process, Raspberry Pi shows the result to the Android phone via Bluetooth. The ECG graph will be shown on the mobile display by pressing the graph button of the application.

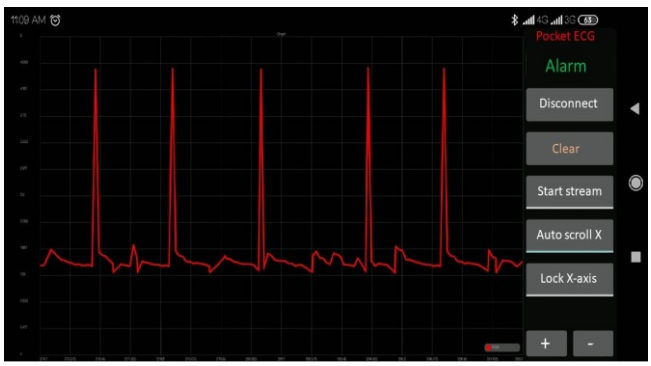

Fig. 5. Real-time ECG graph in Android application

The deep learning structure is executed by Python programming language using the Keras library with a

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TensorFlow backend in Raspberry Pi 3. This structure is trained for 1000 epochs along with a batch size of 5. Table I shows all the hyperparameters use for the deep learning structure. This experiment conducts several patients' ECG signals from the dataset to test the accuracy of each algorithm with LSTM. The additional the epochs, the more the structure will be trained. But it may start over-fitting the training dataset and will be inaccurate with the test dataset.

TABLE I. LSTM MODEL PARAMETERS

Name	Parameters
Hidden layer	32
Input dimension	14999
Epochs	1000
Batch size	5
Function activation	Sigmoid
Calculation of loss	Cross entropy
Optimizer	Adam
Accuracy check	Matrix

After the training procedure, the forecast is made using the features in the testing dataset through the trained model RNN and compares the predictions against the actual values. This research work conducts with the MIT-BIH arrhythmia database, demonstrates a root mean square mistake of 0.004 and a precision of 97.57% on atrial fibrillation arrhythmia detection which is higher than previous research. The overall outcomes show visibly that the arrangement is accurate. For the accuracy, the arrangement's communication overhead, training period, and additional parameters have been optimized. Unlike many deep-learning algorithms, it has small computational costs and meets timing requirements for continuous execution on wearable devices with limited processing power.

TABLE II. PERFORMANCE TABLE

Name	Value
Classification rate for train data	99%
Classification rate for test data	97%
Overall model accuracy	97.57%
Model loss	0.01%
Total root mean square error	0.004

With all these options, the entire component price of this research work is approximately USD 58 or BDT 4900 which is less than previous research. It can be determined that the research is feasible for deployment in developing nations.

V. CONCLUSION

This study has been successfully established an ECG signal classification structure that can forecast the arrhythmia (atrial fibrillation) and able to plot real-time ECG graph. LSTM, RNN, TensorFlow, Keras use here to make a machine learning structure to detect atrial fibrillation which is approximately accurate. Once trained, these models are quite quick and don't require hand-designed characteristics or data preprocessing. Comparing the graph plotted on the android phone, the original graph is seemingly close. The ECG sensor is a three-electrode arrangement. Therefore, complete ECG signals can't capture, but the intervals are easily detected. From the ECG intervals, this system enables to detect atrial fibrillation by using the machine learning model. In the upcoming, some work requirements to be done on noise cancellation for improving the shape of the ECG signal. As a low-cost support decision device with effective features, this arrangement can be an excellent choice for doctors and cardiac patients.

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